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Australia's national security, defence and economic settings right is no easy task. Still, the future of our nation will be increasingly reliant on achieving this kind of aligned outcome.

Smith and Houston have their work cut out for them, especially when it comes to preparedness. If they think beyond the traditional defence lens, they won't be alone in doing so.

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Magnesium market highlights continuing fragility of global supply chains

John Coyne



Image: Paul/Flickr.

Vast and, more often than not, hidden supply and value chains underpin modern Australian life. We've become accustomed to, if not overly confident in, the ability of markets to meet our every need and want. Covid-19 and a series of concurrent natural disasters have left us with a nagging feeling that things might not really be so rosy. It's become painfully apparent that some countries won't always act in the interest of open global trade.

Addressing this challenge requires some big, new policy thinking. Global magnesium supply chains illustrate how difficult that will be.

The supply and value chains for magnesium are a long way from the minds of average Australians. Many would assume that because Australia is a bulk mineral ore exporter without a significant manufacturing base, there's no need for magnesium to be a concern. That assumption is wrong.

Magnesium is a critical input for major and emerging economies' economic and industrial development. It has diverse high-tech applications in a wide range of sectors, from renewable energy to aerospace, defence to transport, and telecommunications to

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agriculture. The global demand for aluminium alloys, made from magnesium, is increasing. Magnesium is critical to the growing market for electric vehicles since its light weight helps to increase their range.

However, for industry and sometimes governments, magnesium supply chains are vulnerable to sudden disruptions.

By 2019, China had become the dominant producer of magnesium, capturing a 94% share of the global export market. Its companies produce magnesium at prices that no other market can match.

Chinese companies have avoided the temptation of using market share to set high prices. Instead, they have set prices that present competitors with high barriers to market entry. The low prices have been a windfall for both manufacturers and consumers, and have helped further entrench China's market dominance.

Interestingly, our understanding of the supply-chain vulnerabilities created by this arrangement wasn't driven by increasing strategic uncertainty but by Chinese domestic policy.

As early as 2006, the Chinese government had become concerned about energy security in light of China's growing economy. In 2016, it introduced a dual-control policy focused on reducing energy intensity and limiting overall consumption. However, the policy wasn't afforded any real priority, so it had a limited impact.

About a year ago, the central government began actively monitoring consumption across China. In September 2021, Shaanxi Province fell victim to the country's 'double control' when it failed to meet energy consumption targets. The government swiftly shut down high-energy-intensity industries, including aluminium production. An international supply crisis ensued, and prices have soared.

Of course, higher prices could make entering the market more attractive to new players. But establishing magnesium value chains—from mines to producing unwrought magnesium and aluminium—takes time. Then there's the added complexity of climate and trade policies. Plus, investors know that the supply chain shortages relate to Chinese domestic policy, which could be reversed just as quickly to prevent market entry.

Canada, the EU, Japan, the UK and the US are all reeling from the sudden disruption of the magnesium and aluminium supply chain. All are trying to find a way to address the issue and reduce prices to avoid further disruption in manufacturing. While markets are part of the solution, they alone won't resolve this problem. Like Australia, most of these countries have limited experience in intervening in markets of this type. A resilient global supply chain will require work from each nation, both individually and in cooperation with one another. Resilience will have a price premium attached to it. Still, this latest episode shows that inaction has a price too.

Australia is an established magnesium producer. Companies like Magneto Metals, with its Batchelor magnesite project in the Northern Territory, are poised to provide supply-chain resilience. Like other companies, Magneto Metals knows that 'to succeed, projects will need to have an integrated low-cost, low-carbon energy strategy and carbon footprint, meet increasingly stringent ESG [environmental, social and governance] policies and a competitive operating cost'.

But unlike similar projects in China, Australian companies face additional costs for infrastructure development, such as roads and affordable power supplies. In the Northern Territory, the future of magnesium rides on investments in facilities like Darwin Harbour's Middle Arm Sustainable Development Precinct. China's market domination still allows it to manipulate pricing, however, and Chinese state-controlled companies have arguably used their market dominance to affect commodity prices in the past in order to inhibit the growth of competitors in the market for rare-earth elements.

A small grouping of like-minded countries could ensure carbon pricing along these supply chains. Such an approach would reward companies that adopt less emissions-intensive processes, resulting in greater competition among suppliers.

Globalisation in one form or another is no doubt here to stay, but the age of predictable, reliable supply chains is over.

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